

Vision Desk AI — Test Knowledge Base

A multi-topic reference document for testing document search and RAG answers

1. Workplace Safety and PPE Requirements

Personal Protective Equipment (PPE) is mandatory on all active construction and industrial sites. Workers must wear a hardhat, safety vest, and steel-toed boots at all times while on site. Sites with active welding or grinding operations additionally require safety goggles and, where noise levels exceed 85 decibels, hearing protection.

In the event of a fall or other injury, the nearest supervisor must be notified immediately and emergency services called if the injury appears serious. The affected area should be cordoned off until it has been inspected. Failure to comply with PPE requirements may result in disciplinary action, including suspension of site access.

Fall protection systems, including guardrails and harnesses, are required for any work performed at a height of six feet or more above a lower level. Scaffolding must be inspected daily by a competent person before each shift begins.

2. Artificial Intelligence Fundamentals

Artificial Intelligence (AI) refers to computer systems capable of performing tasks that typically require human intelligence, such as recognizing patterns, making decisions, and understanding language. Machine Learning (ML) is a subset of AI in which models improve their performance by learning from data rather than following explicitly programmed rules.

Deep Learning is a further specialization within Machine Learning that uses multi-layered neural networks to model complex patterns in large datasets. It underlies most modern computer vision and natural language processing systems, including the object-detection models used for real-time PPE compliance checks.

Common applications of AI include healthcare (disease prediction and diagnostic imaging), finance (fraud detection and credit scoring), transportation (autonomous vehicles), and e-commerce (recommendation systems). The quality of an AI system's output depends heavily on the quality and diversity of its training data; biased or incomplete data tends to produce biased or unreliable predictions.

3. Computer Vision and Object Detection

Object detection models identify and localize objects within an image or video frame, typically returning a bounding box, a class label, and a confidence score for each detected object. YOLO (You Only Look

Once) is a widely used family of real-time object detection architectures that processes an entire image in a single forward pass, making it well suited to live video analysis.

In a PPE compliance system, an object detector is typically trained to recognize classes such as Hardhat, Safety-Vest, and Person, alongside negative classes such as NO-Hardhat and NO-Safety-Vest, which flag workers who are missing required equipment. Confidence thresholds are used to filter out low-certainty detections that are more likely to be false positives.

Video-based detection typically works by sampling frames at a fixed interval rather than analyzing every frame, since consecutive frames are highly similar and full-frame analysis would be computationally expensive. Results are then aggregated across sampled frames to produce a summary of violations.

4. Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation combines a search step with a language-generation step. When a user asks a question, the system first retrieves the most relevant chunks of text from a knowledge base using vector similarity search, then passes those chunks to a large language model as context so it can generate an answer grounded in the retrieved material rather than relying purely on what it memorized during training.

This approach reduces hallucination and allows a system to answer questions about content it has never been trained on, such as a company's internal documents. The quality of a RAG system depends on both the quality of the retrieval step (finding the right chunks) and the quality of the generation step (writing an accurate answer from those chunks).

Chunking strategy matters: documents are typically split into overlapping segments of a few hundred words so that relevant information isn't lost at chunk boundaries, and so that each chunk is small enough to fit comfortably within a language model's context window alongside several other retrieved chunks.

5. Site Emergency Procedures

In case of fire, all personnel must evacuate immediately via the nearest marked exit and proceed to the designated assembly point. Do not use elevators during an evacuation. A headcount will be taken at the assembly point by the site safety officer, and no one may re-enter the building until authorized by emergency services.

For medical emergencies, the site's first-aid kit is located at the main site office. Only personnel with current first-aid certification should administer aid beyond basic first response. All incidents, regardless of severity, must be logged in the incident report system within 24 hours.